ModelIt!

Student Activity Pack

The Water Crisis: Who Gets the Water?

When Rain Runs Out and People Keep Moving In

Grade 05L2 | NGSS 5-ESS2-2, 5-ESS3-1 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Where does the water in your faucet come from before it reaches your house?

2 What do you think happens underground when it rains?

3 If a town grows from 1,000 people to 10,000 people, what happens to water supply?

4 What does the word 'drought' mean, and have you ever experienced one?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Rainfall Amount | Population Size | Groundwater Level | Water Usage Rate | Water Availability

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Conditions

Moderate rainfall and population

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought + Growth

Rainfall at 20%, Population at 80%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Recovery

Restore rainfall to 60% while keeping population high

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Desalination — Converting ocean water to fresh water using technology

Agricultural Irrigation — Water used to grow crops, often the largest water consumer

Water Recycling — Treating and reusing wastewater for non-drinking purposes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What happens when a growing community runs out of water? Use evidence from your simulation to support your answers.

1. **Why are some communities more vulnerable to water crises than others?**

2. **What surprised you about how quickly groundwater can be depleted?**

3. **How does the water crisis model connect to what you know about the water cycle?**

4. **If you could add one component to this model to make it more realistic, what would it be?**

5. **What responsibility do individuals have to conserve water?**

8 STEM Challenge: Design a Water Conservation Plan

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YOUR ENGINEERING MISSION

Design a plan that keeps a growing town from running out of water during a drought.

THE SCENARIO

The town of Dry Creek has doubled in population over 10 years, but rainfall has decreased by 30%. The mayor has asked your team to create a water conservation plan.

Engineering Design Thinking

1. WHAT WATER CONSERVATION RULES WILL YOU PUT IN PLACE?
2. HOW WILL YOU CAPTURE AND STORE RAINWATER WHEN IT DOES RAIN?
3. CAN YOU RECYCLE GREYWATER (USED SINK/SHOWER WATER) FOR IRRIGATION?
4. HOW WILL YOU MEASURE WHETHER YOUR PLAN IS WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Hydrologists and Water Resource Engineers study how water moves through the environment and design systems to ensure communities have enough clean water. They earn $70,000-$110,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the water crisis model, what happens during the Drought + Growth scenario (Rainfall at 20%, Population at 80%)?

- **A.** Groundwater Level stays stable because the aquifer is deep
- **B.** Groundwater Level drops rapidly because water usage far exceeds rainfall refill
- **C.** Water Availability increases because people conserve during droughts
- **D.** Population decreases automatically to match the water supply

2 What type of relationship exists between Population Size and Water Availability?

- **A.** Positive: more people means more water is available
- **B.** No relationship: population does not affect water
- **C.** Negative: more people competing for the same supply reduces water availability per person
- **D.** Positive: more people build more wells, increasing supply

3 Why does groundwater recovery take much longer than groundwater depletion?

- **A.** Because water evaporates before it can reach the aquifer
- **B.** Because rainfall seeps into aquifers slowly over time, while pumping can drain them quickly
- **C.** Because aquifers are destroyed when they run dry and cannot refill
- **D.** Because rain only falls during summer months

4 A town's population triples while rainfall stays the same. Based on the model, what is the most likely result?

- **A.** Nothing changes because rain provides unlimited water
- **B.** Water Usage Rate triples, groundwater depletes, and water availability drops sharply
- **C.** The aquifer automatically expands to meet the demand
- **D.** Rainfall increases to match the population

5 Which solution would the model predict is most effective for a town facing both drought and population growth?

- **A.** Building more wells to pump water faster
- **B.** Waiting for the drought to end naturally
- **C.** Reducing water usage through conservation while exploring new supply sources like water recycling
- **D.** Moving the town to a location with more rain